

Slide 1 - Title

KijaniKiosk end-to-end delivery

Brian Kagai Macharia · Track B · Moringa DevOps Capstone

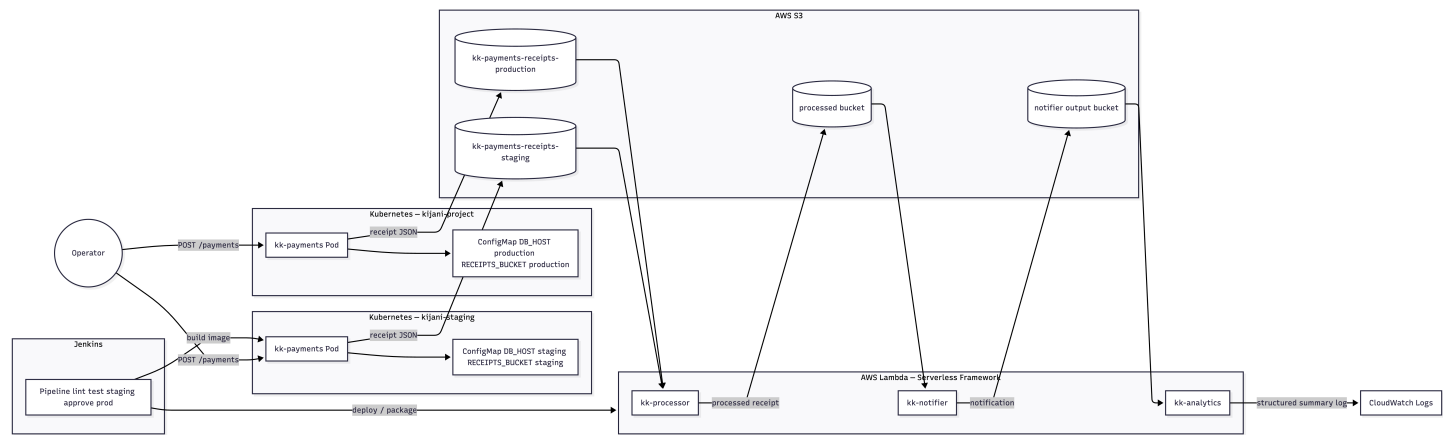
Slide 2 - Problem & scope

- Problem: Payments in K8s had no automated receipt ? analytics path
- Track B: Serverless chain + K8s producer + Jenkins governance
- In scope: 3 Lambdas, kk-payments?S3, CI/CD with approval gate
- Out of scope: Multi-region, managed RDS

Slide 3 - Architecture

!Architecture

- Labelled flows: POST /payments ? S3 ? processor ? notifier ? analytics
- Staging vs production namespaces and buckets



Slide 4 - Key technical decision

Decision: Kustomize base + overlays instead of duplicate Deployment files

Option Trade-off

Duplicate YAML per env Simple but drifts on probe/image changes
Kustomize overlays (chosen) One Deployment; patches for ConfigMap only
Helm chart Heavier for two environments

Slide 5 - Pipeline demo

- Merge to main ? lint/test ? package serverless
- Staging deploy (or offline package when SKIP_AWS_DEPLOY=true)
- Smoke: npm test + receipt fixture
- Approval gate with required reason ? production

Screenshot: Jenkins input step

Slide 6 - AI tooling & governance

- Cursor Agent for serverless + K8s bridge
- Log: docs/ai-governance-log.md (3 entries)
- Example mistake: Lambda folder named kk-payments - caught in review
- Checklist: docs/governance-checklist.md

AWS account pending Complete registration; SKIP_AWS_DEPLOY=false
Static AWS keys in K8s IRSA / OIDC provider
Analytics lists entire bucket DynamoDB aggregate or SQS batch
No Prometheus in cluster yet Apply observability/prometheus/alerts/

Slide 8 - Peer feedback & outcomes

- 3+ peer issues logged (docs/peer-feedback-log.md)
- validate-all.sh added from peer feedback
- Tests: 3 serverless + kk-payments unit tests

Slide 9 - Live demo checklist

1. ./scripts/validate-all.sh
2. Jenkins build (approval gate)
3. _(When AWS ready)_ upload-test-receipt.sh + CloudWatch
4. _(When cluster ready)_ smoke-k8s-payment.sh

Slide 10 - Thank you / Q&A

Repository: <https://github.com/MachariaBrian12/kijani-kiosk>

Tag: v1.0.0 (create before final submit)